Project on **Variational Control for Guidance in Diffusion Models**

MVA - Generative Modeling for Images

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1 Introduction

Diffusion models have emerged as a premier class of generative models, demonstrating exceptional sample quality and the ability to serve as powerful generative priors for a variety of downstream tasks. These models operate by learning to reverse a gradual noising process, effectively mapping simple Gaussian noise back to the data manifold. However, a significant challenge remains in guiding these models to satisfy specific external constraints—such as those found in inverse problems or style-conditioned generation—without requiring extensive retraining or task-specific modifications.

The dominant paradigms for steering diffusion processes are classifier guidance and classifier-free guidance. While effective, these methods often necessitate training additional noise-conditioned models or retraining the base diffusion model for every new conditioning task. To address these limitations, training-free methods like Diffusion Posterior Sampling (DPS) [1] were developed. DPS approximates the diffusion posterior by using Tweedie's formula to estimate the clean data x_0 at each timestep and then applying a gradient-based update to satisfy a terminal cost. Despite its flexibility, DPS relies on simplistic Gaussian approximations that can lead to slow convergence, high sensitivity to hyperparameters, and a lack of stability in complex non-linear tasks.

In this work, we explore the framework of Diffusion Trajectory Matching (DTM) [2], which revisits the guidance problem through the lens of variational inference and stochastic optimal control. Unlike prior methods that treat guidance as a simple additive gradient, DTM models guided diffusion as a controlled Markov process. By treating control signals as variational parameters and optimizing them to satisfy a terminal cost Φ while remaining close to the unguided trajectory, DTM provides a principled unification of various guidance techniques. A key instantiation of this framework is Non-linear Diffusion Trajectory Matching (NDTM), which enables state-of-the-art performance on linear, non-linear, and blind inverse problems using pre-trained pixel or latent-space models without any additional training.

2 Methods

2.1 Algorithm steps

The core idea of the proposed algorithm is to frame diffusion guidance as a stochastic optimal control problem. Instead of simply modifying the score at each step, the authors introduce a control signal $\mathbf{u}_t$ that steers the pre-trained diffusion trajectory toward a target state while remaining close to the original sample manifold.

The Variational Objective and KL Regularization: To find the optimal control $\mathbf{u}_t$, we define a cost function $\mathcal{C}(\mathbf{u}_t)$ at each timestep t. This objective consists of two competing terms:

1. A **Terminal Cost (Φ)** that Encourages the final generated image $\mathbf{x}_0$ to satisfy external constraints. For example, in the case of linear non blind inverse problems, we have explicitly: $\Phi(\hat{\mathbf{x}}_0) = \frac{1}{2}||\mathbf{A}\hat{\mathbf{x}}_0 - \mathbf{y}||^2$ where A is a linear operator (a blurring operator for example).
2. A **Transient Cost ($\mathcal{C}_{tr}$)** that acts as a regularizer using the Kullback-Leibler (KL) divergence to ensure that the guided posterior $p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}_t)$ does not deviate too far from the unguided posterior $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$.

So, the local cost function at time t is formulated as follows:

$$\mathcal{C}(\mathbf{u}_t) = w_T \Phi(\hat{\mathbf{x}}_0^t) + D_{KL}(p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}_t)||q(\mathbf{x}_{t-1}|\mathbf{x}_t)) \tag{1}$$

where $\hat{\mathbf{x}}_0^t$ is the current estimate of the clean image (Tweedie's estimate) and w_T is a weighting scalar.

Gaussian Simplification and KL Computation: In the context of standard diffusion models, both the guided and unguided posteriors are modeled as Gaussian distributions with the same covariance $\sigma_t^2 I_d$:

- Unguided: $q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \sigma_t^2 I_d)$
- Guided: $p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t + \gamma \mathbf{u}_t, t), \sigma_t^2 I_d)$

where γ is the guidance weight. Because the covariances are identical, the KL divergence simplifies significantly to the squared Euclidean distance between the means of the two Gaussian distributions:

$$D_{KL}(p||q) = \|\mu_\theta(\mathbf{x}_t + \gamma \mathbf{u}_t, t) - \mu_\theta(\mathbf{x}_t, t)\|_2^2 \tag{2}$$

This makes the regularization term easy to compute and differentiate during sampling.

Tractable Upper Bound and Gradient Descent: For samplers like DDIM, the paper derives a tractable upper bound for the objective:

$$\mathcal{C}(\mathbf{u}_t) \leq w_T \Phi(\hat{\mathbf{x}}_0^t) + \kappa_t^2 \|\mathbf{u}_t\|_2^2 + \tau_t^2 \|\epsilon_\theta(\mathbf{x}_t + \gamma \mathbf{u}_t, t) - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2 \tag{3}$$

where κ_t and τ_t are time-dependent coefficients. ϵ_θ is the pre-trained denoiser that basically predicts the amount of noise added to $\mathbf{x}_0$ for a given noisy state $\mathbf{x}_t$ at time t.

This structure directly leads to **Algorithm 1 (NDTM)**: Between each standard DDIM step, the algorithm performs N steps of gradient descent to optimize $\mathbf{u}_t$ based on $\nabla_{\mathbf{u}_t} \mathcal{C}(\mathbf{u}_t)$. Once the optimal $\mathbf{u}_t^*$ is found, the state is updated and the next DDIM step is performed using the guided state $\mathbf{x}_t + \gamma \mathbf{u}_t^*$.

Algorithm 1: NDTM (with DDIM). Sampling proceeds by inferring the control followed by sampling from the guided posterior at any time t

Input: Optimization Steps: N, Guidance: γ, Pretrained denoiser: $\epsilon_\theta(\cdot, \cdot)$, Timestep schedule: $\{t\}_{j=0}^T$, DDIM Coefficients: α_t, σ_t, Loss Weights: τ_t, κ_t, w_T

Initialization: $\mathbf{x}_T \sim \mathcal{N}(0, I_d)$

for $t = T$ ***to*** 1 **do**

- $\mathbf{u}_t^{(0)} = 0$
- $\hat{\epsilon}_{\text{uncond}} \leftarrow \epsilon_\theta(\mathbf{x}_t, t)$
- **for** $i = 0$ ***to*** $N - 1$ **do**
  - $\hat{\epsilon}_{\text{control}}^{(i)} \leftarrow \epsilon_\theta(\mathbf{x}_t + \gamma \mathbf{u}_t^{(i)}, t)$
  - $\hat{\mathbf{x}}_0^{(i)} \leftarrow \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t + \gamma \mathbf{u}_t^{(i)}]$
  - $\mathcal{C}_{\text{score}} = \tau_t^2 \left\| \hat{\epsilon}_{\text{uncond}} - \hat{\epsilon}_{\text{control}}^{(i)} \right\|_2^2$
  - $\mathcal{C}_{\text{control}} = \kappa_t^2 \|\mathbf{u}_t\|_2^2$
  - $\mathcal{C}_{\text{terminal}} = w_T \Phi(\hat{\mathbf{x}}_0^{(i)})$
  - $\mathcal{C}_t^{(i)} \leftarrow \mathcal{C}_{\text{score}} + \mathcal{C}_{\text{control}} + \mathcal{C}_{\text{terminal}}$
  - $\mathbf{u}_t^{(i+1)} \leftarrow \text{Update}(\mathbf{u}_t^{(i)}, \nabla_{\mathbf{u}_t} \mathcal{C}_t^{(i)})$
- $\mathbf{x}_{t-1} \leftarrow \text{DDIM}(\mathbf{x}_t + \gamma \mathbf{u}_t^*, t)$

return $\mathbf{x}_0$

2.2 Comparison with DPS

The proposed Non-linear Diffusion Trajectory Matching (NDTM) generalizes and improves upon classical guidance methods like Diffusion Posterior Sampling (DPS).

The Gradient Descent Loop: Multi-step vs. Single-step: The most prominent difference is the introduction of a nested optimization loop.

- The **DPS** (Classical Guidance) algorithm approximates the conditional score using a single gradient step based on the likelihood of the current Tweedie estimate $\hat{\mathbf{x}}_0$. This approach assumes a linear control where the signal is directly added to the score function. However, this crude approximation often leads to slow convergence and instability.
- In **NDTM** (Proposed), the control signal $\mathbf{u}_t$ is treated as a variational parameter optimized via a loop of N gradient descent steps at each timestep t. This iterative process allows the model to dynamically adjust the trajectory to satisfy the terminal cost $\Phi(\hat{\mathbf{x}}_0^t)$ while strictly adhering to the original manifold via the KL-regularization.

Sampler Dynamics, DDPM vs DDIM: The second fundamental difference between the two approaches lies in the choice of the underlying diffusion sampler, which dictates how the latent space is traversed.

- The original formulation of **DPS** uses **DDPM** (Denoising Diffusion Probabilistic Models [3]): DDPM is defined as a Markovian reverse process where each transition $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is stochastic, involving the injection of Gaussian noise at every step. DPS relies on a first-order approximation of the posterior score. Since it only performs a single gradient update per timestep, it can tolerate (and sometimes benefit from) the stochasticity of DDPM, which helps in exploring the posterior distribution. However, this randomness makes it difficult to perform precise, multi-step trajectory optimization.
- **NDTM** uses **DDIM** (Denoising Diffusion Implicit Models [4]): DDIM generalizes the diffusion process to a non-Markovian setting. By setting the noise parameter to 0, each transition becomes entirely deterministic. Because NDTM performs an inner loop of N gradient descent steps to optimize the control signal $\mathbf{u}_t$, it requires a stable and consistent relationship between the current state and the terminal estimate $\hat{\mathbf{x}}_0^t$. Furthermore, the KL-based regularization derived in the paper specifically leverages the DDIM coefficients to ensure the guided trajectory stays on the data manifold.

The DPS algorithm can still be used with DDIM steps, and this is what is done in the experiments of the paper. This is also what we did in the following experiments in order to guarantee fairness of the comparison of the DPS and NDTM algorithms.

3 Experiments

3.1 Deblurring on FFHQ

We obtained these first results on the deblurring (Gaussian kernel, $\sigma = 3$) task. In this experiment, the terminal cost has the following expression:

$$\Phi(\hat{\mathbf{x}}_0^t) = ||\mathbf{k} * \hat{\mathbf{x}}_0^t - \mathbf{y}||^2$$

where $\mathbf{k}$ is the Gaussian blur kernel and $\mathbf{y}$ is the blurry version of the original image $\mathbf{x}_0$.

For the NDTM algorithm we use the following parameters: $T = 50$, $N = 2$ (optimization steps), $w_T = 50$, $\gamma = 4$ (guidance coefficient).

For the DPS algorithm we use the following parameters: $T = 100$ and $\zeta = 1$.

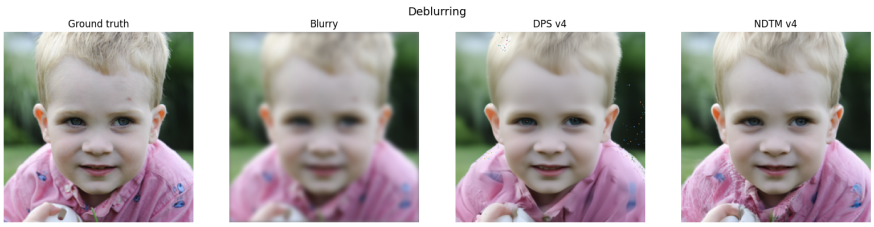


Figure 1: Comparison of deblurring task DPS (PSNR=26.34) vs. NDTM (PSNR=27.55)

To obtain the results of the Figure 2, we selected 5 images from the FFHQ dataset and applied the NDTM and DPS algorithms to the blurred versions of these images. The figure reports the PSNR score for each image and for each algorithm, as well as the average PSNR score of each algorithm over the 5 images.

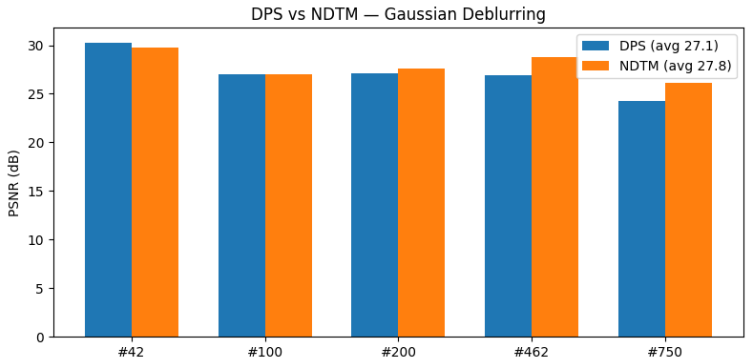


Figure 2: Further evaluation of deblurring task DPS vs. NDTM on a few samples

We see that on average (at least on these examples), the NDTM algorithm performs slightly better than the DPS algorithm.

We can compare the two algorithms by looking at the number of times the denoising model ϵ_θ is evaluated during the deblurring process:

- For the DPS algorithm, at each time step $t \in \{0, 1, ..., T-1\}$ the estimate $\mathbf{x}_t$ is passed through the denoising model ϵ_θ. This results in T evaluations of ϵ_θ.
- For the NDTM algorithm, at each time step $t \in \{0, 1, ..., T'-1\}$ the estimate $\mathbf{x}_t$ is passed through the denoising model ϵ_θ to obtain $\hat{\epsilon}_{\text{uncond}}$. Then for each optimization step $n \in \{0, 1, ..., N-1\}$ the guided estimate $\mathbf{x}_t + \gamma \mathbf{u}_t^{(i)}$ is passed through ϵ_θ to compute $\hat{\epsilon}_{\text{control}}^{(i)}$. This results in $T' \times (1+N)$ evaluations of ϵ_θ.

So in this experiment, we performed 150 evaluations of the denoising model in the NDTM algorithm while we performed only 100 in the DPS algorithm. This observation questions the interest of the NDTM algorithm over DPS.

We can further test the algorithm on a deblurring task with a more complicated blur kernel, for instance a motion blur with a rather highly nonlinear motion. We generated such a kernel with the MotionBlur library.

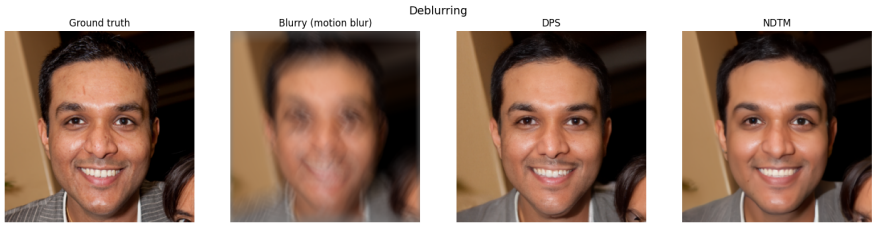


Figure 3: Comparison of deblurring task (motion blur) DPS (PSNR=27.43) vs. NDTM (PSNR=27.90)

We see again on this example that the two algorithms perform almost exactly the same, with a slight advantage for NDTM. The performances of the NDTM algorithm on the linear deblurring task are globally satisfying but we shall keep in mind that the NDTM algorithm that we ran here involved slightly more computations than DPS.

3.2 Non-linear deblurring

For this experiment, we first apply to the image a Gaussian kernel and then apply downsampling (scale factor of 0.125 so we get to an intermediate image of size 32×32) and then apply upsampling, this makes the blurring operation nonlinear.



Figure 4: Comparison of non linear deblurring task DPS (PSNR=25.89) vs. NDTM (PSNR=24.47)

On this example, both algorithms perform a bit worse than in the previous case but are still reasonable.

3.3 Random inpainting on FFHQ

We obtained these results on the random (90% masking) inpainting task. In this experiment, the terminal cost has the following expression:

$$\Phi(\hat{\mathbf{x}}_0^t) = ||\mathbf{M} * \hat{\mathbf{x}}_0^t - \mathbf{y}||^2$$

where $\mathbf{M}$ is the random mask applied to the image and $\mathbf{y}$ is the masked version of the original image $\mathbf{x}_0$.

Here we didn't tune the hyperparameters, we used the same as in the previous task (deblurring).

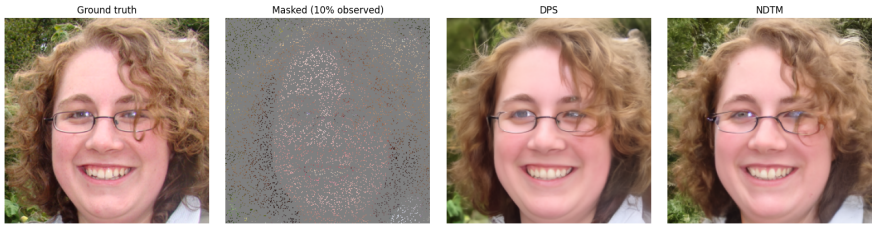


Figure 5: Comparison of random inpainting task DPS (PSNR=24.46) vs. NDTM (PSNR=23.50)

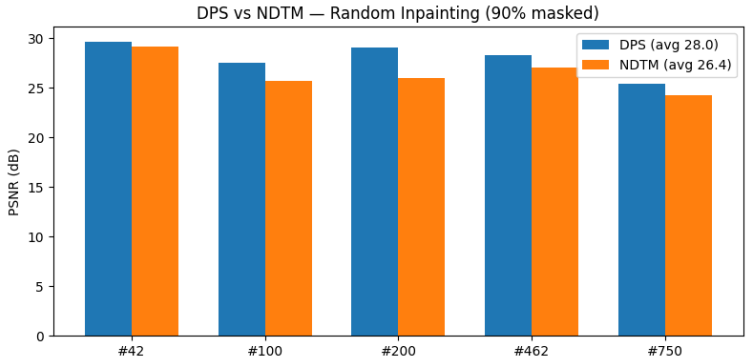


Figure 6: Further evaluation of random inpainting task DPS vs. NDTM on a few samples

To obtain the results of the Figure 6, we used the 5 images from the previous task and applied the NDTM and DPS algorithms to the masked versions of these images. The figure reports the PSNR score for each image and for each algorithm, as well as the average PSNR score of each algorithm over the 5 images.

We see that on average (at least on these examples), the NDTM algorithm performs slightly worse than the DPS algorithm.

3.4 Large area inpainting

This experiment is very similar to the previous one, we just use a predefined mask (a square) that hides a large area of the original image and the terminal cost remains the same.



Figure 7: Comparison of large area inpainting task DPS (PSNR=27.44) vs. NDTM (PSNR=18.95). The NDTM algorithm performs poorly on this example.

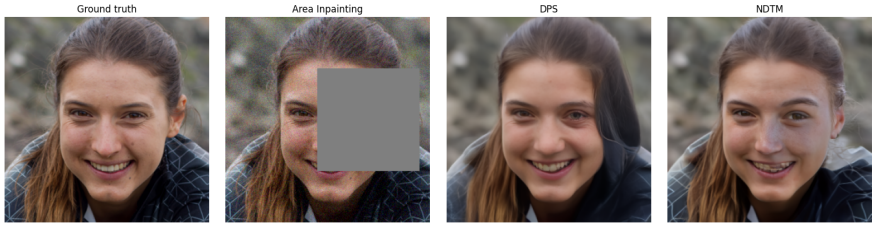
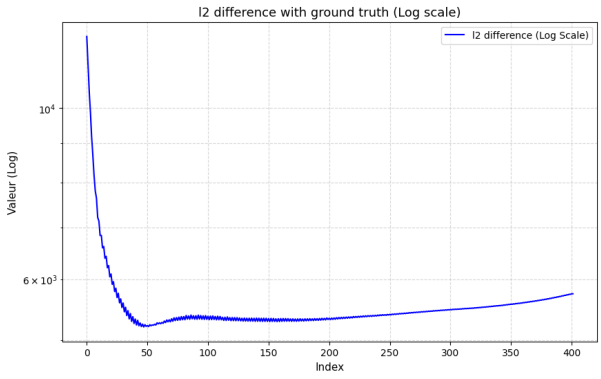
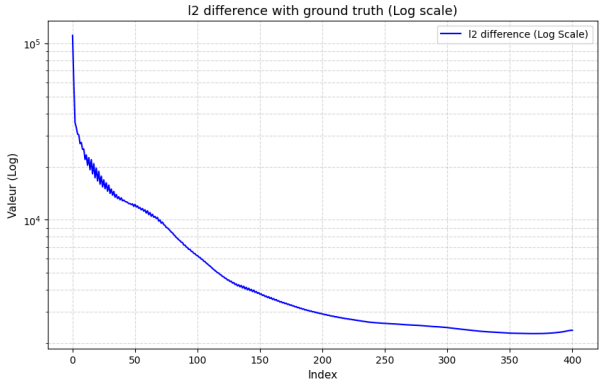


Figure 8: Comparison of large area inpainting task DPS (PSNR=20.73) vs. NDTM (PSNR=22.15). Both algorithms struggle on this image but NDTM performs slightly better in terms of PSNR.

To further analyze the reconstruction performance on the large-area inpainting task, we monitor the L_2 distance between the generated samples and the ground truth across the sampling iterations. As illustrated in Figures (a) and (b) below, both models exhibit a clear performance plateau, indicating that additional computational steps are unlikely to yield significant improvements in reconstruction fidelity. This behavior is especially evident for the NDTM model; its L_2 error curve stabilizes remarkably early in the process. This rapid convergence suggests that the model quickly reaches its optimal state for the given constraints, establishing a hard ceiling on the achievable reconstruction accuracy under the current experimental setup.



(a) L_2 distance between the generated samples and ground truth for NDTM



(b) L_2 distance between the generated samples and ground truth for DPS

3.5 Improvement remarks

The core principle of NDTM is to steer the diffusion process by inferring a control signal that optimizes the approximation of the posterior distribution, as detailed in the original paper.

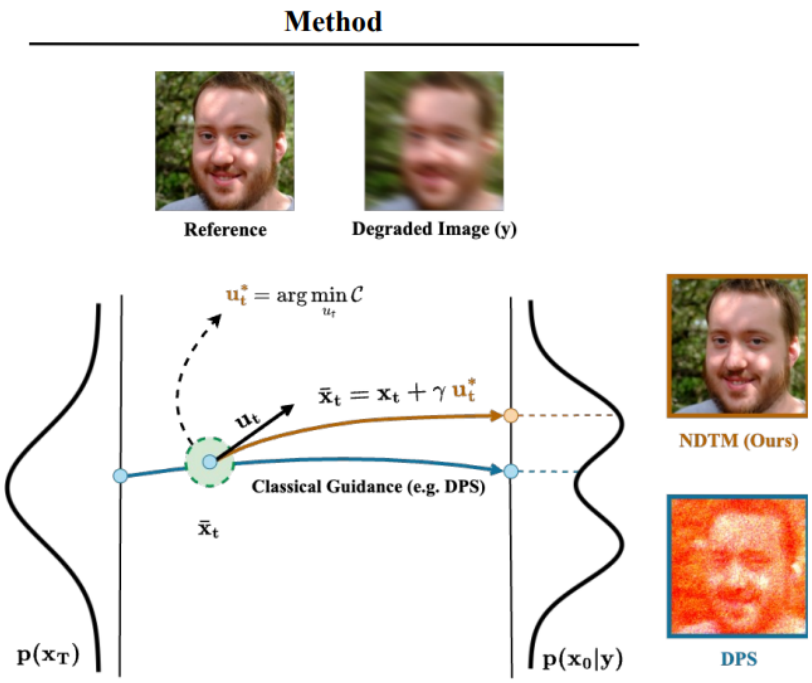


Figure 10: Idea behind NDTM according to the article

However, our empirical findings suggest that it may be more effective to apply directional guidance exclusively during the initial stages of the process before transitioning to a DPS (Diffusion Posterior Sampling) approach.

For instance, in large-area inpainting tasks, NDTM tends to initialize with a suboptimal estimate and subsequently converges toward that specific state. Conversely, by sustaining directional guidance over a longer period, we obtain a more accurate, albeit blurry, approximation of the ground truth. A subsequent blind deblurring step could then potentially yield superior results compared to the standard NDTM or DPS implementations.



Figure 11: Result of only directional guidance for the large area inpainting task

4 Conclusion

This comparative study between **Non-linear Diffusion Trajectory Matching (NDTM)** and **Diffusion Posterior Sampling (DPS)** highlights the persistent challenges of guidance in diffusion models. While the variational control framework proposed by NDTM is theoretically elegant and offers a formal unification of various guidance methods, our empirical findings suggest that its practical advantages remain marginal when compared to the robustness and simplicity of DPS.

In the majority of the linear and non-linear restoration tasks tested, the performance gains (in terms of PSNR and computational time) provided by NDTM were often negligible relative to its implementation complexity. A significant hurdle for NDTM's adoption is its high sensitivity to hyperparameters, specifically the guidance weight γ and the regularization coefficients. Unlike DPS, which remains relatively stable across different scenarios, NDTM requires meticulous and time-consuming tuning for each specific task, making it less intuitive for general-purpose applications.

Our experiments on large-area inpainting were particularly revealing. While DPS struggles to reconstruct coherent structures in large missing regions, NDTM also fails to provide a satisfying alternative, frequently stagnating prematurely on blurry or out-of-distribution solutions. This observation calls into question the utility of the method for highly under-determined inverse problems, where the added complexity of the control signal does not compensate for the lack of local information.

Nevertheless, the concept of integrating control theory into the diffusion process remains a fertile research direction. Our observations suggest that NDTM's current inefficiency may stem from an overly uniform application of control. Performing the same number of optimization iterations at every timestep appears sub-optimal. It seems that directional guidance is most crucial during the initial stages of the reverse process, where the global structure of the image is established. A more promising approach would involve remodeling this control to be adaptive—concentrating computational effort on the early trajectory before transitioning to a lighter guidance method like DPS.

In conclusion, while NDTM provides a solid theoretical foundation, its direct application currently underperforms compared to established standards, necessitating a deeper reflection on the temporal dynamics of control within generative models.

5 Appendices

5.1 DDPM (Denoising Diffusion Probabilistic Models)

Forward process The forward process progressively adds noise to the data:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t;\ \sqrt{\alpha_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}) \tag{4}$$

where $\alpha_t = 1 - \beta_t$ and β_t follows a linear schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

Thanks to the Markov structure, we can sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big) \tag{5}$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{6}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

Tweedie estimate (prediction of $\mathbf{x}_0$ from $\mathbf{x}_t$)

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}} \Big(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\ \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \Big) \tag{7}$$

Reverse process (DDPM sampling)

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1};\ \boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\ \tilde{\beta}_t \mathbf{I}\big) \tag{8}$$

where the posterior mean is:

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\ \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) \tag{9}$$

and the posterior variance is:

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\ \beta_t \tag{10}$$

The sampling step is:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\tilde{\beta}_t}\ \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{11}$$

Training objective

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} \left[\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \|^2 \right] \tag{12}$$

5.2 DDIM (Denoising Diffusion Implicit Models)

DDIM defines a non-Markovian reverse process that shares the same marginals $q(\mathbf{x}_t \mid \mathbf{x}_0)$ as DDPM but allows deterministic sampling with fewer steps.

Generalized reverse step Given a subsequence of timesteps $\tau_1 < \tau_2 < \cdots < \tau_S \subseteq \{1, \ldots, T\}$, the DDIM update from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ (where t and $t-1$ refer to consecutive elements in the subsequence) is:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\ \underbrace{\hat{\mathbf{x}}_0(\mathbf{x}_t)}_{\text{predicted } \mathbf{x}_0}\ + \underbrace{\sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\ \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}_{\text{direction pointing to } \mathbf{x}_t} + \underbrace{\sigma_t\ \mathbf{z}}_{\text{noise}} \tag{13}$$

where:

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\ \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{14}$$

The noise scale σ_t interpolates between DDPM and deterministic:

$$\sigma_t = \eta \sqrt{\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}} \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \tag{15}$$

- $\eta = 0$: fully deterministic (DDIM)
- $\eta = 1$: recovers DDPM stochasticity

Key properties

- Uses the **same pretrained model** ϵ_θ as DDPM — no retraining.
- Can use far **fewer steps** (e.g. 50 instead of 1000) by subsampling the timestep schedule.
- The $\bar{\alpha}_t$ values are looked up from the **original** $T = 1000$ schedule at the selected subsequence indices.

5.3 DPS (Diffusion Posterior Sampling)

Problem setup Given a degradation model:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma_y \, \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{16}$$

we want to sample from the posterior $p(\mathbf{x}_0 \mid \mathbf{y})$.

Conditional score decomposition Using Bayes' rule on the score:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t \mid \mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{y} \mid \mathbf{x}_t)}_{\text{likelihood score}} \tag{17}$$

DPS approximation The intractable likelihood $p(\mathbf{y} \mid \mathbf{x}_t) = \int p(\mathbf{y} \mid \mathbf{x}_0) \, p(\mathbf{x}_0 \mid \mathbf{x}_t) \, d\mathbf{x}_0$ is approximated using a Dirac delta at the Tweedie estimate:

$$p(\mathbf{x}_0 \mid \mathbf{x}_t) \approx \delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0(\mathbf{x}_t)) \tag{18}$$

This gives:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} \mid \mathbf{x}_t) \approx -\nabla_{\mathbf{x}_t} \left\| \mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0(\mathbf{x}_t)) \right\|^2 \tag{19}$$

where the gradient flows through the Tweedie estimate $\hat{\mathbf{x}}_0$ back to $\mathbf{x}_t$ via the chain rule.

DPS sampling step Combining the unconditional DDIM step with the guidance gradient:

$$\mathbf{x}_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}} \, \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \, \epsilon_\theta(\mathbf{x}_t, t) + \sigma_t \mathbf{z}}_{\text{standard DDIM step}} - \underbrace{\zeta_t \frac{\nabla_{\mathbf{x}_t} \left\| \mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0) \right\|^2}{\left\| \nabla_{\mathbf{x}_t} \left\| \mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0) \right\|^2 \right\|}}_{\text{DPS guidance (norm-normalized)}} \tag{20}$$

where ζ_t is the guidance step size.

5.4 Summary of relationships

	DDPM	**DDIM**	**DPS**	**NDTM**
Stochastic	Yes ($\tilde{\beta}_t$)	Controlled by η	Inherits from DDIM	Inherits from DDIM
Steps needed	$T = 1000$	$S \ll T$ (e.g. 50)	S (100–1000)	S (e.g. 50)
Guidance	None	None	$-\zeta_t \nabla_{\mathbf{x}_t} \Phi(\hat{\mathbf{x}}_0)$	$\min_{\mathbf{u}_t} \mathcal{C}(\mathbf{u}_t)$
Inner loop	No	No	No	Yes (N steps)
Model	$\epsilon_\theta(\mathbf{x}_t, t)$	Same	Same	Same
Markovian	Yes	No	No	No

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